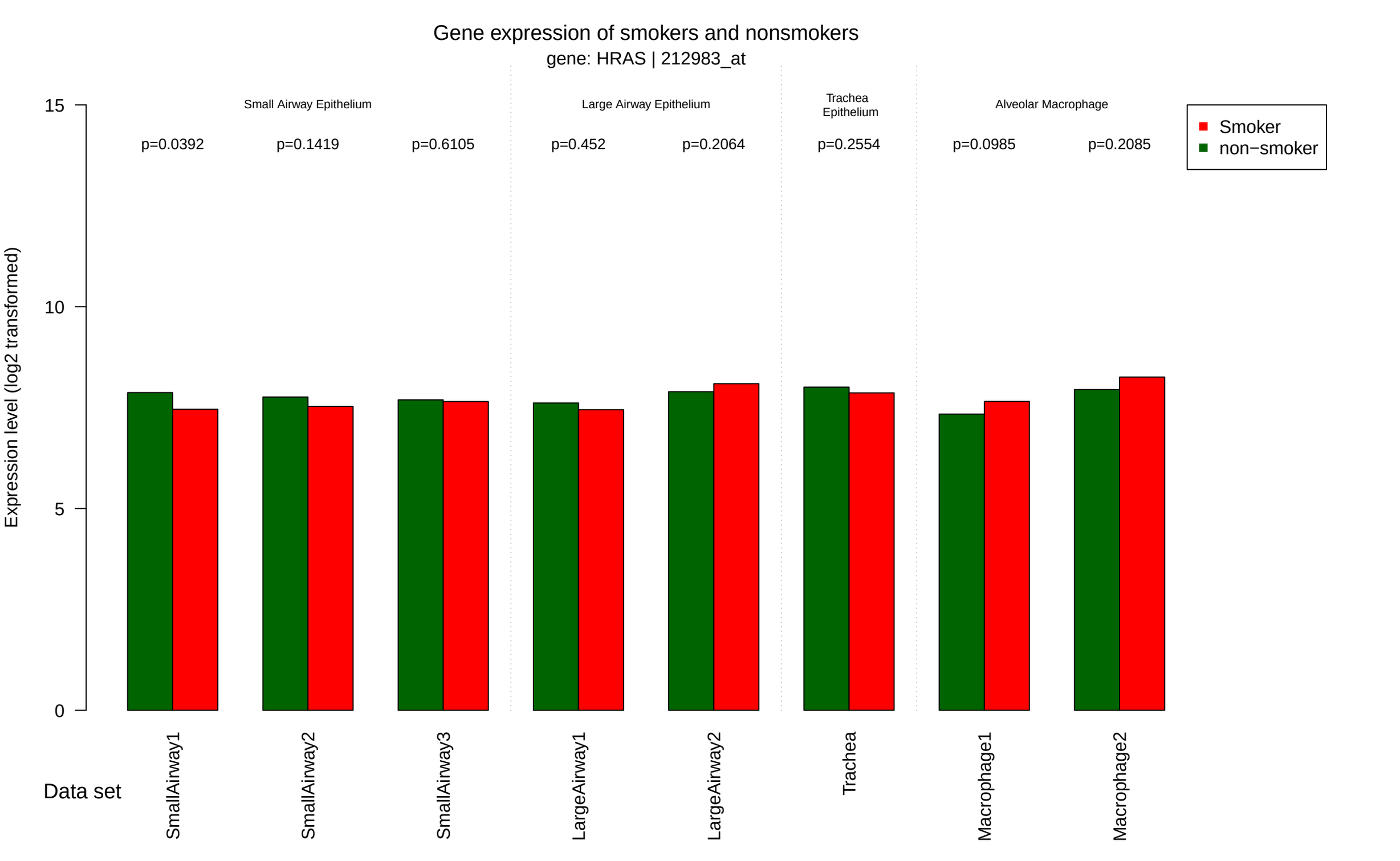
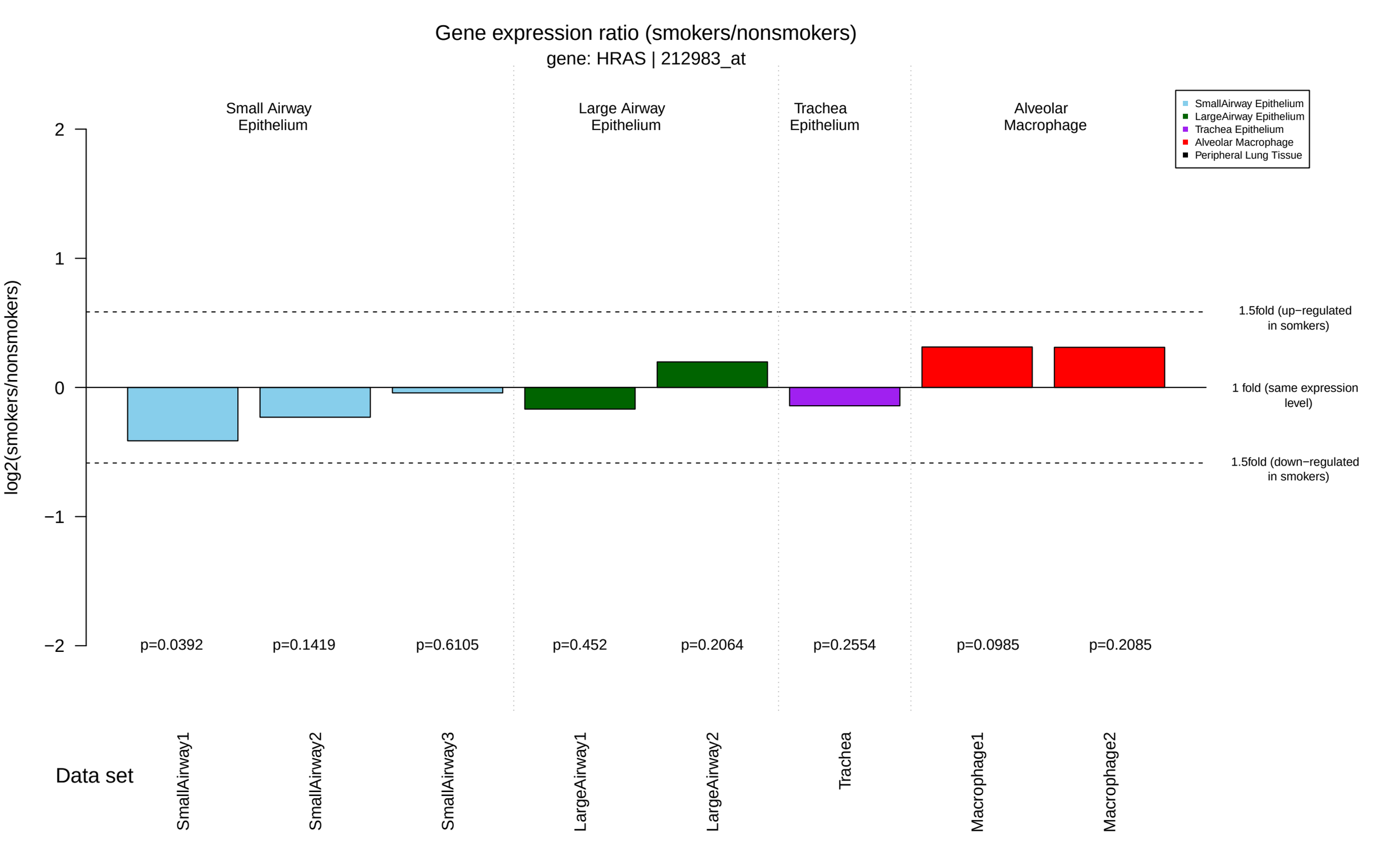
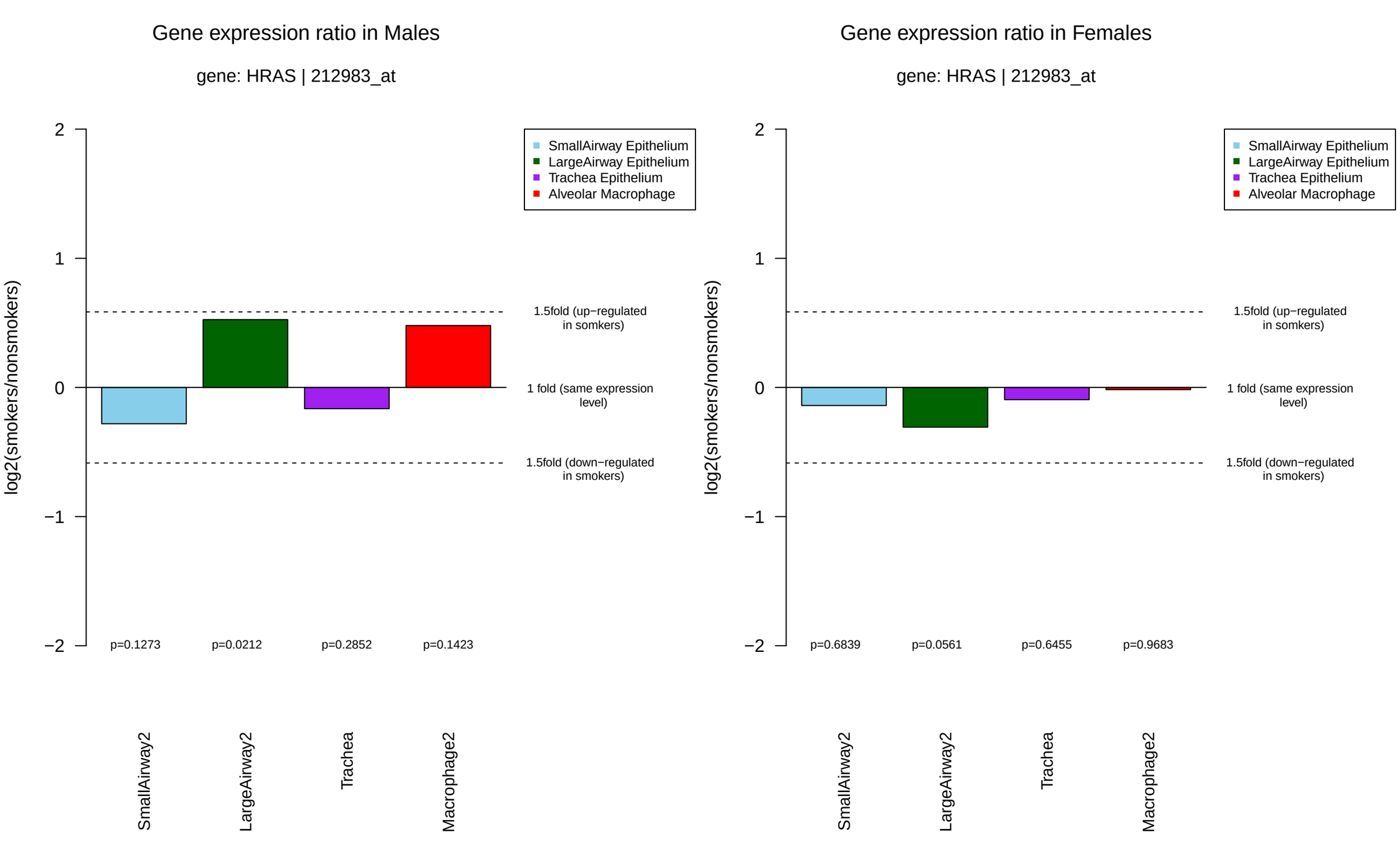
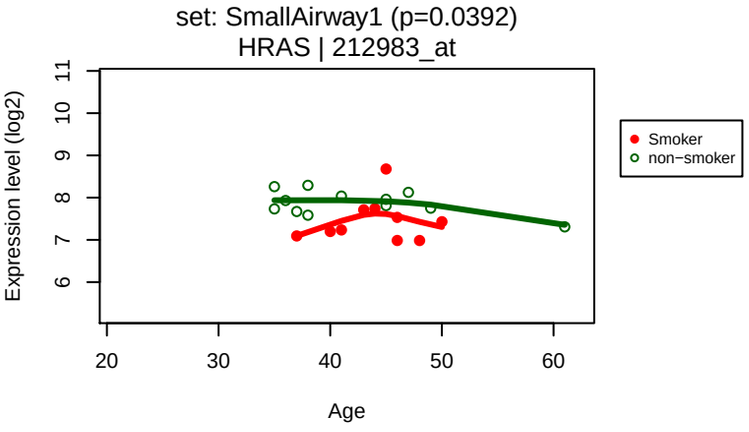
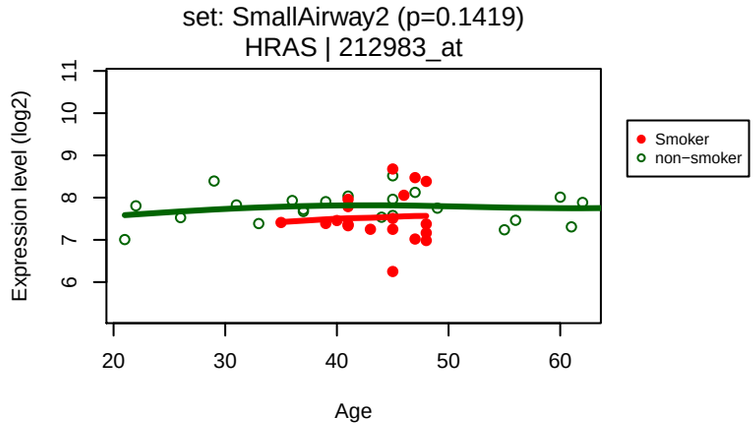
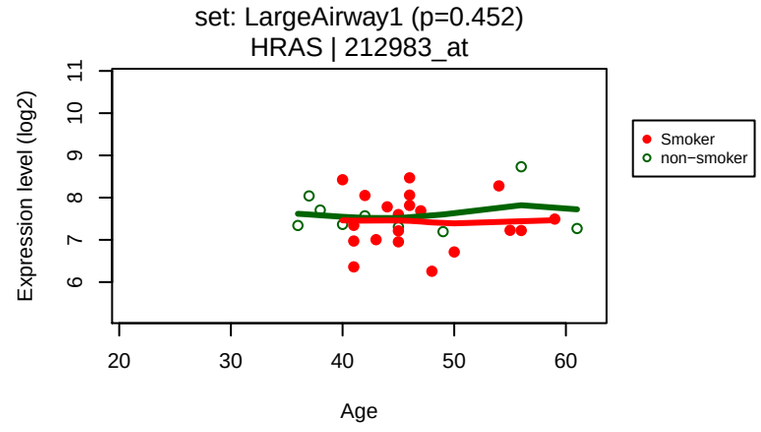

Gene expression of smokers and nonsmokers
gene: HRAS | 212983_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0392
p=0.1419
p=0.6105
p=0.452
p=0.2064
p=0.2554
p=0.0985
p=0.2085
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

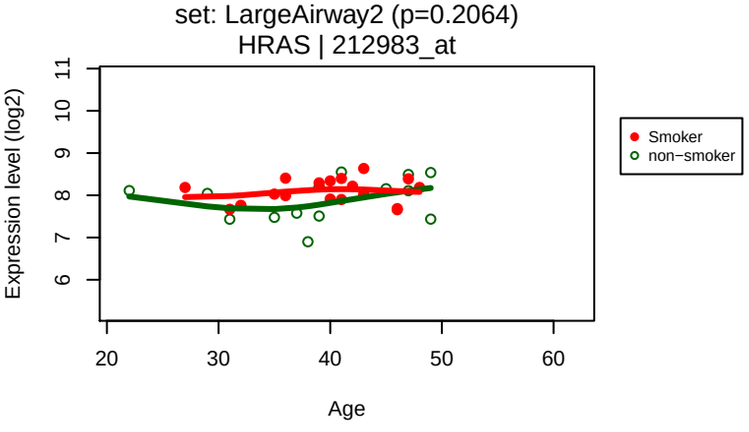

Gene expression ratio (smokers/nonsmokers)
gene: HRAS | 212983_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0392
p=0.1419
p=0.6105
p=0.452
p=0.2064
p=0.2554
p=0.0985
p=0.2085
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

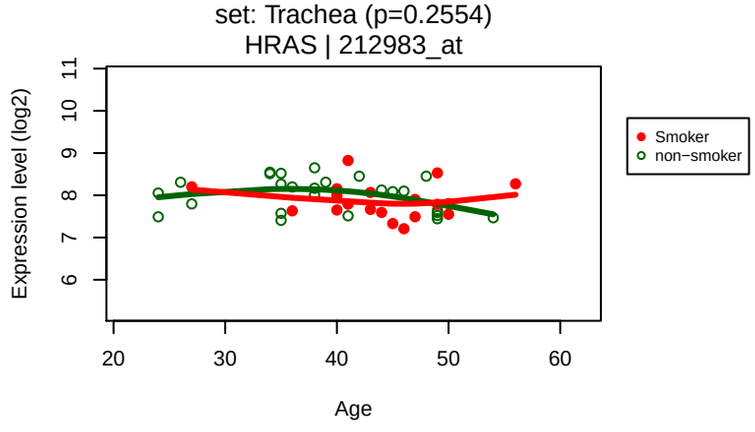

Gene expression ratio in Males
gene: HRAS | 212983_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1273
p=0.0212
p=0.2852
p=0.1423
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HRAS | 212983_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6839
p=0.0561
p=0.6455
p=0.9683
SmallAirway2
LargeAirway2
Trachea
Macrophage2

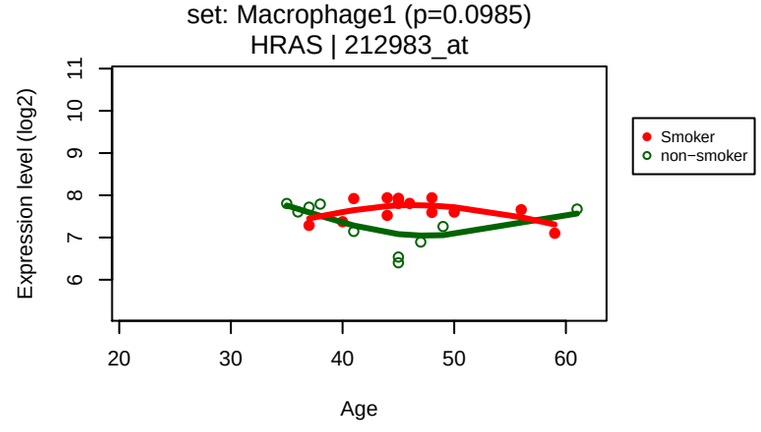

set: SmallAirway1 (p=0.0392)
HRAS | 212983_at
Expression level (log2)
Age
Smoker
non−smoker

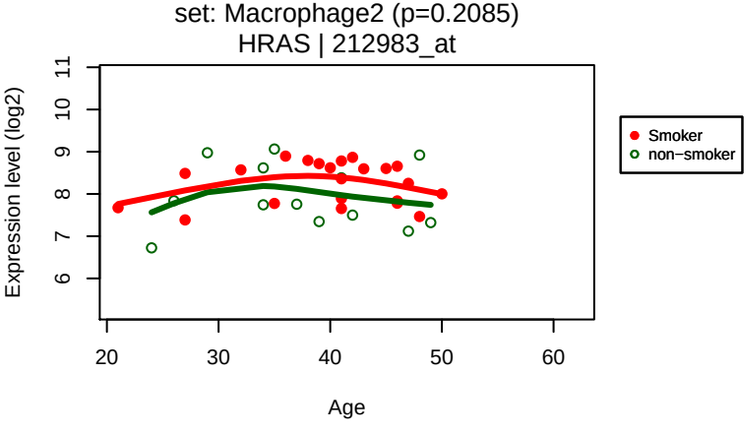

set: SmallAirway2 (p=0.1419)
HRAS | 212983_at
Expression level (log2)
Age
Smoker
non−smoker

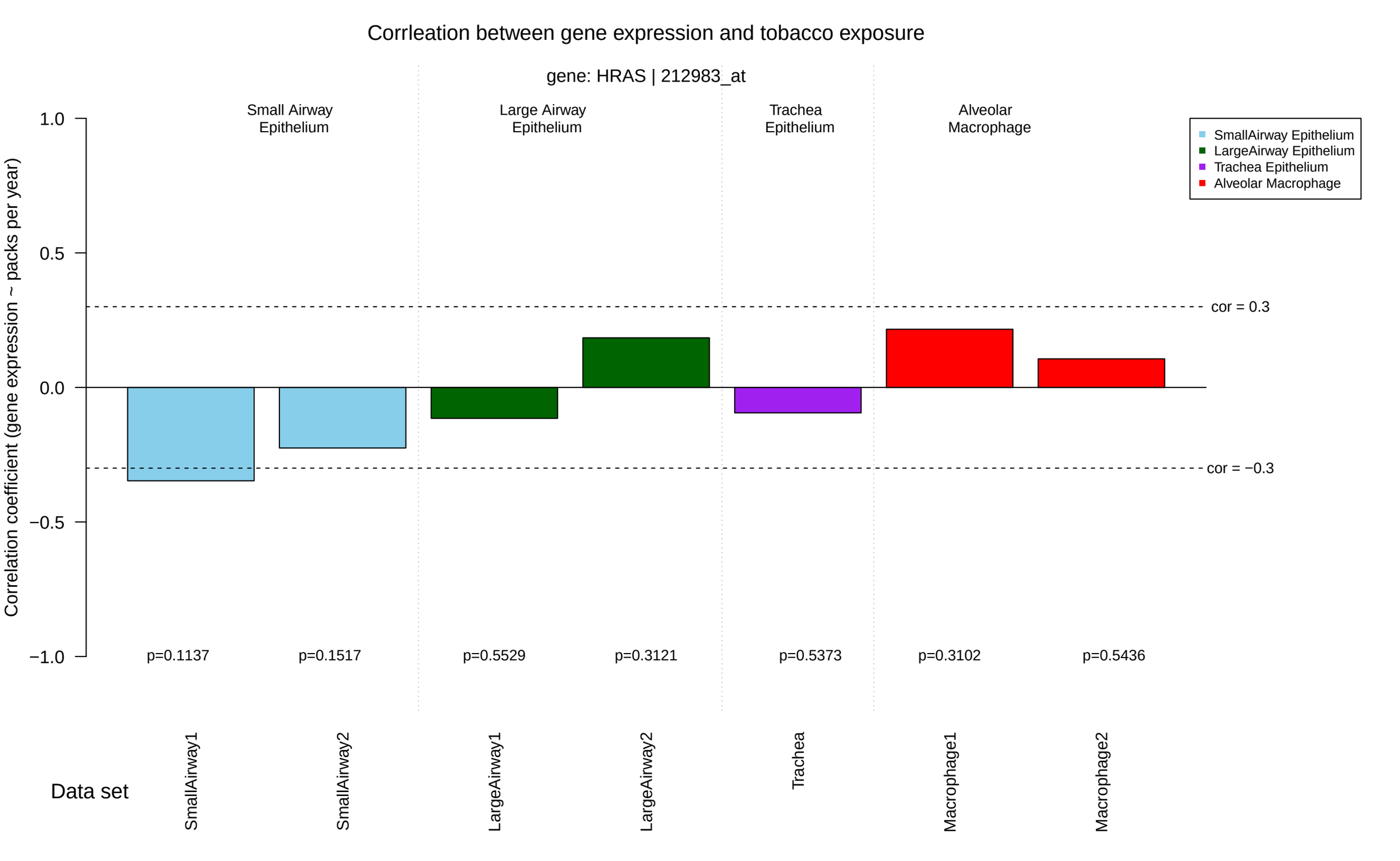

set: LargeAirway1 (p=0.452)
HRAS | 212983_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2064)
HRAS | 212983_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2554)
HRAS | 212983_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0985)
HRAS | 212983_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2085)
HRAS | 212983_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HRAS | 212983_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1137
p=0.1517
p=0.5529
p=0.3121
p=0.5373
p=0.3102
p=0.5436
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2